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OF SCIENCE AND TECHNOLOGY
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Group Project
Course Title: Software Processes

Course Code: SPS611S

1. Project Overview

In this project, you will apply software engineering principles and follow a structured software development process from inception to testing. You will work in a team to develop a small software system while documenting all major phases of the Software Development Life Cycle (SDLC). A central focus of the project is selecting, applying, and justifying a **software process model** that fits your chosen problem and context

2. Learning outcomes

By the end of this project, you should be able to:

- Analyse a realistic software problem and define a clear problem statement, objectives, and scope.
- Select an appropriate software process model and justify its suitability for a given context.
- Produce structured artefacts for requirements, design, implementation, and testing.
- Apply collaborative development practices (version control, task management, documentation).
- Present and demonstrate a working prototype and reflect on how your chosen process worked in practice

INSTRUCTIONS

GROUP Project (6 members per group)
Due Date: 29 March to 26 April 2026 (***Refer to Submission format and logistics***)
Due Time: 23H55PM

<u>Projects Structure</u>
Submission:
Written Report (8–12 pages)
Presentation (5 –10 slides, 10 minutes, all members must participate).
Total Marks: 100 Marks

Project Topics
Each group must select one problem requiring a software-based solution. The problem must be relevant to Namibia, NUST, or broader societal challenges and must be approved by the lecturer before proceeding.

Table 1: Topics to select from

	Topic	Possible clients
1	Campus Facility/Venue Booking System – A system that allows staff to view and book available campus facilities.	NUST Facilities department
2	Digital Mental Health Support Platform for NUST Students and staff. A system providing anonymous check-ins, self-help resources, and counselling booking.	NUST Department of Student Services NUST Staff NUST Students
3	Marginalised learners often face high dropout rates. Many struggle with a lack of motivation and guidance. A need for a Digital Role Model Platform to Support Marginalised Boarding School Learners.	Marginalised learners (San, Ovatue, Ovatjimba, and Ovazemba) School Staff Marginalised Communities Ms Kays
4	Digital Class/Lesson Attendance Tracking System for NUST Classes.	NUST Lecturers NUST Students
5	Automated Grouping System for NUST First-Year Students.	NUST Lecturers NUST Students
6	A Secure and Trustworthy Online Dating System.	Professionals, Mature individuals, Safety-Conscious Women, LGBTQ+ Community
8	Smart Navigation System for the NUST Campus.	NUST Facilities department
9	Digital Internship Management System for NUST Students	NUST Lecturers NUST Faculties NUST Coporative Education Unit Department
10	Namibian AI & Computational Thinking Activity Hub (ML-for-kids focus)	Teachers & Ms Muntuumo
11	Digital Role Model & STEM Activity Platform for Marginalised Learners	Teachers, Role modelsb. & Ms Muntuumo
12	Mood tracking system.	
13	IilongaSTEM: Sustainable STEM Activity Builder (5-Step Co-Design Platform)	Teachers & Ms Muntuumo

5. Software process requirement

Each group must select, apply, and document a suitable Software Development Process Model (for example: Waterfall, Iterative, Scrum, Kanban, Spiral, DevOps or a justified hybrid).
You must:

- Describe the chosen model clearly in your own words.
- Justify why this model is appropriate for your project, considering:
    - Project size and complexity.
    - Stakeholder involvement and availability.
    - Team skills, tools, and time constraints.
- Provide concrete process artefacts consistent with the chosen model, such as:

    - For Agile/Scrum: product backlog, sprint backlog, sprint schedule, retrospective notes, screenshots of Kanban boards.
    - For Waterfall/Iterative: phase plan, milestones, Gantt-style timeline, phase review notes.

- Show evidence that you followed the process, including:
    - How does your actual work compare to your initial plan?
    - Changes/adaptations you made along the way.
    - A short reflection on what worked, what did not, and what you would improve.

The software process model and its application are a major focus of this assignment and carry a significant portion of the marks.

6. Required deliverables

You must submit the following deliverables.

1. Project Proposal
    - Context and background.
    - Problem statement and motivation.
    - Objectives and a clearly defined scope.
    - Main stakeholders.
2. Software Process Model Description and Justification
    - Description of the selected process model.
    - Justification of its suitability for your project.
    - Process artefacts (plans, backlogs, Kanban board screenshots, meeting notes, etc.).
    - Short reflection on how the process worked in reality.
3. Software Requirements Specification (SRS)
    - Functional requirements.
    - Non-functional requirements.
    - Assumptions and constraints.
4. System Design
    - Use case diagram(s).
    - High-level architecture diagram.
    - Database design (ERD or schema).
5. Prototype
    - UI wireframes or a clickable prototype illustrating main user flows and key screens.
6. Implementation (Working System)
    - Implementation of core functionality to support key use cases.
    - Source code managed through a version control system.

7. Testing Report
    - Test cases (inputs, expected outputs).
    - Test execution results (pass/fail, screenshots if needed).
    - Brief discussion of discovered defects and fixes.
8. Final Presentation and Demonstration
    - 5–20 slides.
    - 10-minute presentation and live demonstration of the system.
    - All group members must participate in the presentation.

7. Submission format and logistics

- Written report: 15+ pages (excluding cover page, references, and appendices).
- Presentation: 5–20 slides, 10 minutes per group.
- Due date: 26 April 2026
- Due time: 23:55
- Submission platform: E-Learning Management System (via Turnitin).

The project contributes 30% to your final semester mark. The report must clearly list all group members, including their student numbers, first names, and last names.

Deliverable	Due Date	Due Time
Project Proposal	29 March 2026	23H55
Software Process Selection & Justification	29 March 2026	23H55
Requirements Specification	5 April 2026	23H55
System Design	5 April 2026	23H55
Prototype	5 April 2026	23H55
Implementation	19 April 2026	23H55
Testing	19 April 2026	23H55
Final Presentation & Demo	26 April 2026	23H55

8. Tools and resources (recommended)

You are encouraged to use:

- IDEs: Visual Studio Code, IntelliJ IDEA, NetBeans, or similar.
- Databases: MySQL, PostgreSQL, SQLite, or similar.
- Version control: Git with GitHub, GitLab, or Bitbucket.
- Design and modelling: Figma, Adobe XD, Balsamiq, StarUML, Lucidchart, Draw.io.
- Project management: Trello, Jira, Monday.com, or similar (for example, Kanban boards for Agile teams).
- Testing: pytest/unittest (Python), JUnit (Java), Jest/Mocha (JavaScript), or similar.

9. Academic integrity

All sources must be cited using APA referencing. AI tools may be used responsibly to support your work (for example, brainstorming ideas or improving language), but they may not replace your own analysis, design, or coding.

Plagiarism is a serious offence and will be dealt with in accordance with NUST policy. Similarity thresholds and penalties (as checked by Turnitin) typically follow:

- 0–20% similarity: 0 marks deducted.
- 21–40% similarity: 10% of project marks deducted.

- 41–60% similarity: 25% of project marks deducted.
- 61–100% similarity: 100% of project marks deducted.

It is your responsibility to be familiar with official NUST policies as published in the Prospectus/Yearbook.

Project Marking Rubric (Total: 100 Marks)

Criteria	**Description**	**Marks**
Project Proposal	Clear problem definition, objectives, and scope.	10
Software Process Selection & Justification	Appropriate choice of software process model with clear justification.	15
Requirements Specification	Well-documented functional and non-functional requirements.	15
System Design	Use cases, architecture diagrams, and database design.	10
Prototype	Clear interface designs demonstrating system functionality.	10
Implementation	Working system demonstrating core functionality.	15
Testing	Evidence of unit/system testing and documented results.	10
Documentation Quality	Clarity, structure, academic writing, and referencing.	5
Final Presentation & Demo	Clear explanation, system demonstration, and ability to answer questions.	10
TOTAL		100

********** **End of Project Document** **********